

Question	Answer	Marks	Guidance
5(a)(i)	complete loop enclosing the origin	B1	There must be an attempt to draw curves and not just straight lines joining the intercepts
	loop cutting x-axis at $x = \pm 2.5$ cm and never at larger magnitudes of x	B1	Allow tolerance $\pm \frac{1}{2}$ small square horizontally.
	loop cutting v-axis at $v = \pm 4.4$ cm s ⁻¹ and never at larger magnitudes of v	B1	Allow tolerance $\pm \frac{1}{2}$ small square vertically from 4.5. (2 small squares from 5.0 = 4.5) <u>Special Case:</u> Allow 2/3 for half-loop that passes through three of the four axis intercepts within tolerance. Annotate cross, tick, tick horizontally.
5(a)(ii)	$v_0 = \omega x_0$ and $\omega = 2\pi f$	C1	Use of r , R or A for x_0 not sufficient unless defined or clarified by the substitution.
	frequency = $4.4 / (2.5 \times 2\pi) = 0.28$ Hz	A1	Full substitution and answer needed. Unit not needed but any unit given must be correct.
5(b)(i)	(oscillation of object at) maximum amplitude	B1	Allow 'vibration' for 'oscillation'.
	when driving frequency = natural frequency (of system)	B1	Allow 'platform' or 'motor' or 'forced' or 'external' for 'driving'. Allow 'ball' for 'system'. Allow 'matches' for '=' Allow 'nature' for 'natural' annotate with BOD
5(b)(ii)	line from $f = 0$ with amplitude never zero when $f > 0$, rising continuously to a single peak and then falling continuously until $f = 0.40$ Hz	B1	Allow zero amplitude everywhere except at a single spike (but not zero amplitude anywhere if peak is not shown as a single spike).
	(single) peak shown at $f = 0.28$ Hz	B1	Allow tolerance ± 1 small square horizontally. No ECF from 5(a)(ii) .